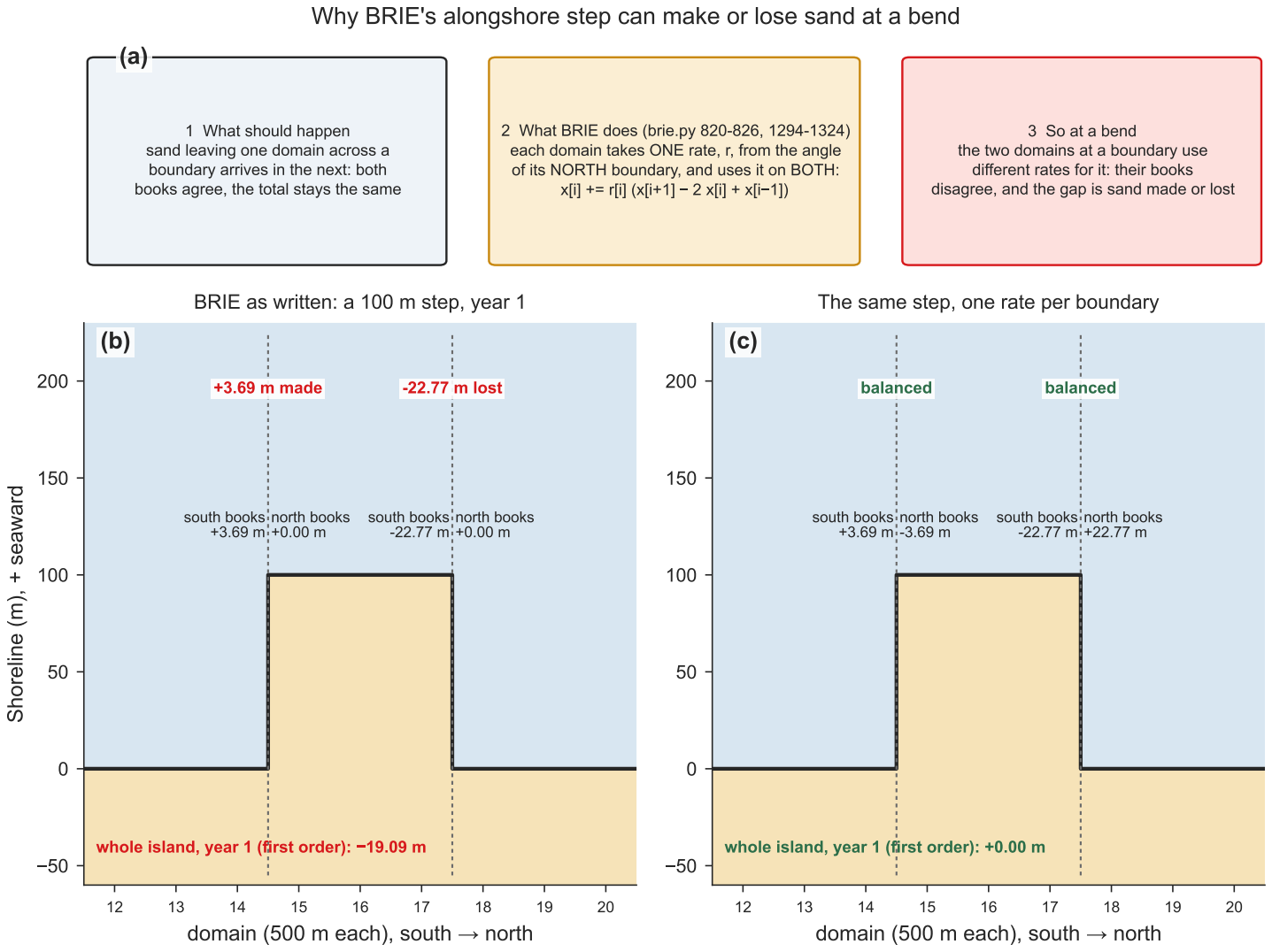
Why BRIE's alongshore step can make or lose sand at a bend
(a)
1 What should happen
sand leaving one domain across a
boundary arrives in the next: both
books agree, the total stays the same
2 What BRIE does (brie.py 820-826, 1294-1324)
each domain takes ONE rate, r, from the angle
of its NORTH boundary, and uses it on BOTH:
x[i] += r[i] (x[i+1] − 2 x[i] + x[i−1])
3 So at a bend
the two domains at a boundary use
different rates for it: their books
disagree, and the gap is sand made or lost
BRIE as written: a 100 m step, year 1
(b)
+3.69 m made
-22.77 m lost
south books +3.69 m
north books +0.00 m
south books -22.77 m
north books +0.00 m
whole island, year 1 (first order): −19.09 m
The same step, one rate per boundary
(c)
balanced
balanced
south books +3.69 m
north books -3.69 m
south books -22.77 m
north books +22.77 m
whole island, year 1 (first order): +0.00 m
Shoreline (m), + seaward
200
150
100
50
0
−50
12
13
14
15
16
17
18
19
20
domain (500 m each), south → north